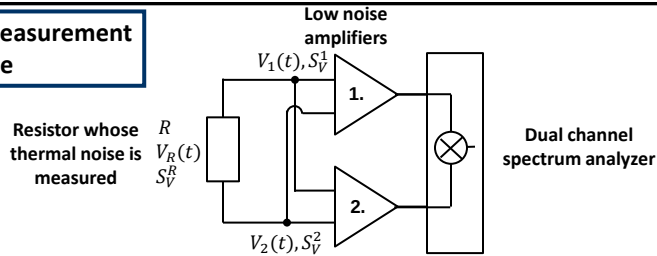


Cross-correlation measurement technique



The resistor's voltage fluctuates over time due to thermal noise: $V_R(t)$

The input noise of the amplifiers ($V_1(t)$ és $V_2(t)$) is added to the thermal noise of the resistor. The input noise of the two amplifiers is independent of each other! Like this, the two amplifiers output $A(V_R(t)+V_1(t))$ and $A(V_R(t)+V_2(t))$ voltage, respectively.

The dual-channel spectrum analyzer calculates a cross-correlation, i.e. it examines the spectrum of the product of two signals! As an illustration of the the method, let's just look at the expected value of the cross-product:

$$\langle A(V_R(t) + V_1(t)) \cdot A(V_R(t) + V_2(t)) \rangle = A^2 \underbrace{\langle V_R^2(t) \rangle}_{S_V^R \cdot BW} + \underbrace{A^2 \langle V_R(t) \rangle \langle V_2(t) \rangle + A^2 \langle V_1(t) \rangle \langle V_R(t) \rangle + A^2 \langle V_1(t) \rangle \langle V_2(t) \rangle}_0$$

If only one amplifier were used, the output voltage would have the mean squared deviation:

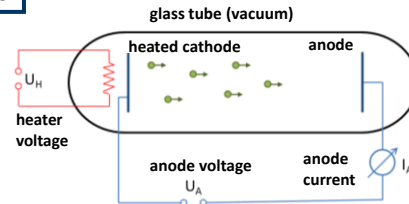
$$\langle A^2 (V_R(t) + V_1(t))^2 \rangle = A^2 \underbrace{\langle V_R^2(t) \rangle}_{S_V^R \cdot BW} + A^2 \underbrace{\langle V_1^2(t) \rangle}_{S_V^1 \cdot BW}$$

That is, for a single amplifier, the sum of the noise density of the sample and the input noise density of the amplifier is measured, while for a cross-correlation of two amplifiers, the noise of the amplifiers is cancelled out. When measuring noise, the amplifier input noise is usually comparable to or greater than the noise under test.

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Shot noise of independently emitted electrons

The shot noise is related to the quantized electron charge, which results in current of individual "electron shots" instead of a constant noiseless current. Schottky made the first shot noise measurement in 1918 measuring the current noise of a vacuum tube. Electrons emitted randomly from a heated cathode are driven by the applied voltage to the anode, where a current is detected. In this case the electrons are emitted randomly and independently of each other, so the electron incidences are well described by the Poisson distribution:



$$P_N(t) = \frac{t^N}{\tau^N N!} e^{-t/\tau}$$

← The probability that N electrons are emitted within time t, such that the average time between the independent electron emissions is τ

This formula can be proved by induction

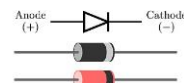
$$\left[P_N(t + dt) = P_{N-1}(t) \frac{dt}{\tau} + P_N(t) \left(1 - \frac{dt}{\tau} \right) \Rightarrow \frac{dP_N(t)}{dt} = \frac{P_{N-1}(t) - P_N(t)}{\tau}, \dots \right]$$

A special property of the Poisson distribution: the mean value of N for a certain time is the same as its mean squared deviation:

$$\langle N \rangle = \sum N \cdot P_N(t) = \frac{t}{\tau}, \quad \langle (\Delta N)^2 \rangle = \langle N^2 \rangle - \langle N \rangle^2 = \langle N \rangle$$

Note, that $\langle (\Delta N)^2 \rangle$ is proportional to the current noise, whereas $\langle N \rangle$ is proportional to the mean current, i.e. the $S_I \sim I$ relation is expected for the shot noise of independently emitted electrons ("Poisson noise")

Independent electron emissions are not only available in a vacuum tube, but also in other systems, like a semiconductor diode at small enough currents.



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$r?$
 window:
 $t/2$

Mathematically, we convolve our signal with a function $g(t)$, which is a Δt wide filtering window:

We have seen earlier that the Fourier transform of a product is equal to the convolution of the Fourier transforms. It can be shown that the Fourier transform of the convolution is the product of the Fourier transforms:

$$S_{\text{out}}(\omega) = S_{\text{in}}(\omega)|g(\omega)|^2$$

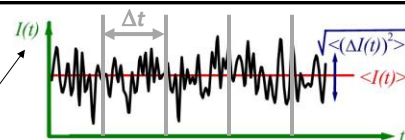
For the same amount of output noise, what rectangular filter of width $\Delta\omega = 2\pi\Delta f$ would be needed assuming white input noise with S_0 density? This is the so-called **equivalent noise bandwidth, ENBW**.

A graph showing the magnitude squared of the transfer function, $|g(\omega)|^2$, versus angular frequency ω . The y-axis is labeled $|g(\omega)|^2$ and has a tick mark at 1. The x-axis is labeled ω . A blue rectangular pulse is shown, starting at $\omega = 0$ and ending at $\omega = \Delta\omega$, with a height of 1. A blue arrow labeled $\Delta\omega = 2\pi\Delta f$ indicates the width of the pulse. A red curve, representing the squared magnitude of the Fourier transform, is shown. It starts at 1 at $\omega = 0$, decreases to a minimum value, and then exhibits a series of decaying oscillations as ω increases.

Similar to the Nyquist-Shannon sampling theorem: at Δt sampling the signal is resolved up to $f_{\max}=1/2\Delta t$. But here we are not sampling at Δt intervals, rather we smooth the signal with a Δt filtering window. This acts as an antialiasing filter suppressing the spectrum above $f_{\max}=1/2\Delta t$.

$$ENBW = \Delta f = \frac{1}{2\Delta t}$$

Independent emission of electrons: Poisson noise



Where N is described by the Poisson distribution:

$$P_N(\Delta t) = \frac{(\Delta t)^N}{\tau^N N!} e^{-\Delta t/\tau}$$

According to the definition of noise density:

- If we measure using Δt interval averaging, this is our bandwidth

$$S_I = 2e\langle I \rangle$$

- This measurement was first carried out by Schottky in vacuum diodes (1918).
- You can try this measurement using a semiconductor diode in the “Measurement techniques laboratory course” together with the determination of the “k” Boltzmann constant by thermal noise measurement.

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Resistance fluctuations, 1/f-type noise

In a field-biased transistor, the gate dielectric may have charge traps present. An electron can be trapped in or released from a charge trap with a characteristic time τ , causing a resistance fluctuation with a correlation function of characteristic time τ :

$$C_R(\Delta t) = \langle \Delta R(t) \cdot \Delta R(t + \Delta t) \rangle = \langle \Delta R^2(t) \rangle \cdot \exp(-\Delta t / \tau)$$

From this the density of the resistance

noise is obtained by Fourier

transformation (Lorentzian noise spectra):

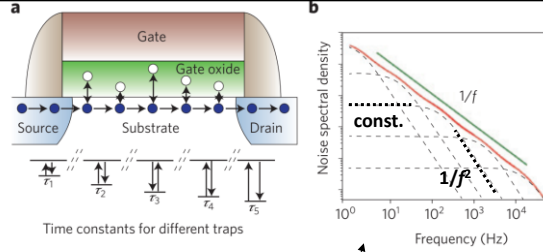
$$S_R(\omega) = 2C_R(\omega) = \frac{4 \langle \Delta R^2(t) \rangle \tau}{1 + \omega^2 \tau^2}$$

Adding together the Lorentzian noise spectra of many different processes with different τ , we get a summed spectrum with shallower frequency dependence than $1/f^2$, the so-called 1/f-type noise. This 1/f-type noise is present in many systems, even for completely different physical reasons, and often causes serious difficulties in measurements. For example, before the introduction of GaN/AlGaN high bandwidth semiconductor devices in communication technologies, years of research and development were needed to reduce the 1/f noise of these devices by about 5 orders of magnitude. Note, that it is not necessarily a truly $1/f$ frequency dependence, rather $1/f^\gamma$ with γ being close to unity.

If voltage noise is measured at constant current, then $\Delta V = I \cdot \Delta R$:

$$S_V(f_0) = \langle (\Delta V(t | f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot \langle (\Delta R(t | f_0, \Delta f))^2 \rangle / \Delta f = I^2 \cdot S_R$$

That is, since 1/f noise is basically resistance noise due to resistance fluctuations, the related voltage noise scales by the square of the driving current, according to Ohm's law!



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Aliasing problem in noise measurements, antialiasing filter

The DFT evaluates the spectrum within the 0 and $\omega_{\max} = 2\pi / (2\Delta t)$ interval at discrete, $2\pi k / (N\Delta t)$ circular frequencies. What happens, if components with circular frequency higher than $\omega_{\max}$ are also present in the signal?

Let us consider a pure harmonic signal:

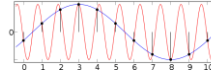
$$F(t) = A_0 e^{i\omega_0 t}, \quad f(\omega) = A_0 2\pi \delta(\omega - \omega_0)$$

Now let's see how the spectrum changes in practice when calculating the DFT. It is important to note that a data acquisition card or oscilloscope basically samples at Δt intervals for $\ll \Delta t$ times, and it is not the case that it averages the signal over Δt times!

$$DFT(\omega_k) = \sum_{n=0}^{N-1} A_0 e^{i\omega_0 n \Delta t} W(n \Delta t) e^{-i\omega_k n \Delta t} \Delta t \quad \text{notice, that: } DFT(\omega_k) = DFT\left(\omega_k + \frac{2\pi m}{\Delta t}\right)$$

where m is an integer, and furthermore: $DFT(\omega_k) = DFT^*(-\omega_k) \rightarrow |DFT(\omega_k)| = |DFT(-\omega_k)|$

Thus, it can be seen that any signal with a high circular frequency is seen as if it were a signal in the $[0, \omega_{\max} = \frac{2\pi}{2\Delta t}]$ domain with its own amplitude A_0 . This is the phenomenon of aliasing.



The components in the $(2m) \frac{2\pi}{2\Delta t} < \omega < (2m+1) \frac{2\pi}{2\Delta t}$, $m = 1, 2, \dots$ intervals appear at the $\omega_m = \omega - \frac{2\pi m}{\Delta t}$

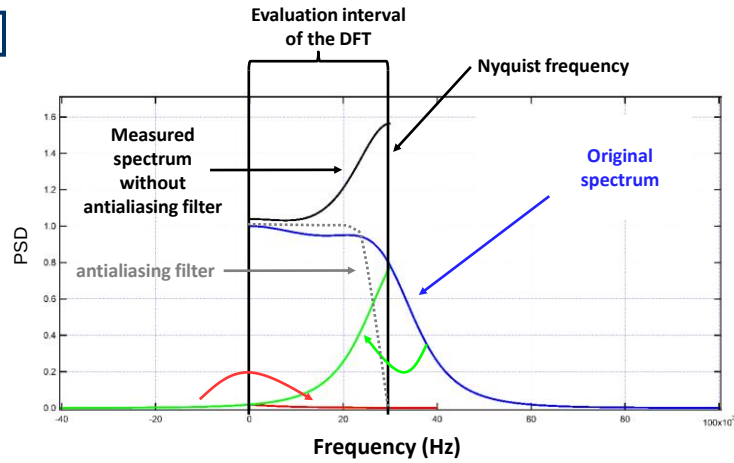
circular frequencies, while the $(2m-1) \frac{2\pi}{2\Delta t} < \omega < (2m) \frac{2\pi}{2\Delta t}$, $m = 1, 2, \dots$

interval converts to $\omega_m = \frac{2\pi m}{\Delta t} - \omega$ circular frequencies.

When measuring noise, we see a continuous spectrum, and each frequency component is in independent phase, so the power density of the frequency components converted from high frequency due to aliasing, i.e. the square of their Fourier transformed absolute value, is added to the real noise density to be measured at a given frequency!!!

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Antialiasing filter



If the measured noise also contains components above the Nyquist frequency ($f_{\max} = 1/2\Delta t$), the aliasing phenomenon causes the components above $f_{\max}$ to be “folded” into the frequency range of the DFT causing a completely false signal. Solution: the signal should be measured with an anti-aliasing filter which cuts off right at the Nyquist frequency, removing the higher frequency components, which would cause aliasing.